SPE-229250-MS

An Integrated Approach to Virtual Flow Metering of Wells with ESP Based on a Combined Hydraulic and Electric Model

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 DOI 10.2118/229250-MS

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

Accurate real-time flow rate estimation for ESP-lifted wells remains challenging due to limited direct measurements, high water cut, gas-oil ratio variations, and pump degradation. This paper presents an integrated virtual flow metering (VFM) approach that combines hydraulic and electrical models of ESP systems within a single physically consistent framework. The algorithm performs multi-stage adaptation — first to hydraulic parameters (intake pressure) and then to electrical telemetry (motor power, amperage, voltage, and load) — followed by optimized weighting of calibration coefficients for production rate recovery.

The method was validated on a large set of wells in Western Siberia. The integrated VFM achieved a mean absolute percentage error (MAPE) of 4,866%, improving accuracy by 1,549% relative to purely hydraulic models. These results demonstrate that accounting for both hydraulic and electrical subsystems enable reliable, high-resolution flow estimation without installing additional downhole metering devices, providing an efficient tool for production optimization and ESP performance management. Flow rate restoration is performed using multi-parameter optimization of a function that considers intake pressure, power, current, voltage, and submersible motor load.

Introduction

In the context of oil production from hard-to-recover reserves, emerging technological challenges require comprehensive consideration of the specific characteristics of unconventional resource development [1], [2], [3]. One such challenge is virtual flow metering — a method used to measure and predict fluid production in oil wells.

In Western Siberia, most of the well stock is equipped with various types of electric submersible pumps (ESPs) [4]. These pumps are complex and expensive systems that require monitoring and specialized maintenance to ensure optimal operating conditions and avoid production losses. Therefore, wells equipped with ESP require a virtual flowmeter — a physical model capable of considering numerous criteria and pump operating parameters to accurately estimate flow rates [5].

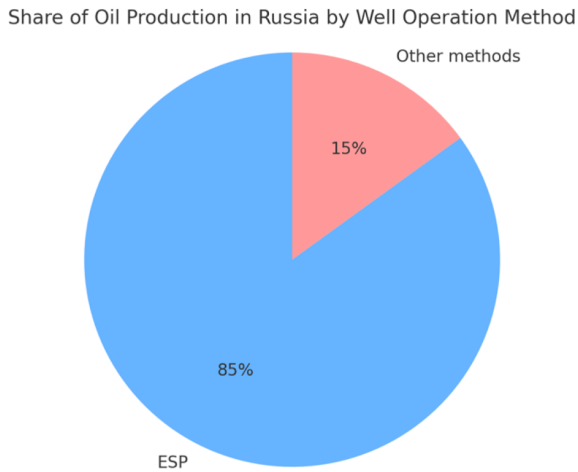

Figure 1—Statistics of Producing Wells.

During oil well operation, data are collected at varying frequencies. The main production indicator — oil output — may only be measured once every few days, while pump operating parameters (e.g., consumed active power) may be recorded in information systems several times per hour, or even several times per minute [6]. The low frequency of production rate measurements is largely due to specific field operation conditions. Oilfields are often vast in size, and production volumes are low due to the late stage of development. These fields have been in operation for many years and are equipped with outdated infrastructure, which typically lacks flow measurement devices on each well.

Moreover, at present, a key challenge for oil companies is the operation of producing oil wells under conditions with a large number of complicating factors, such as low temperatures, high gas factors, gas breakthroughs, and so on [7], which makes the monitoring of well performance and the quantitative assessment of its operating mode an extremely high priority.

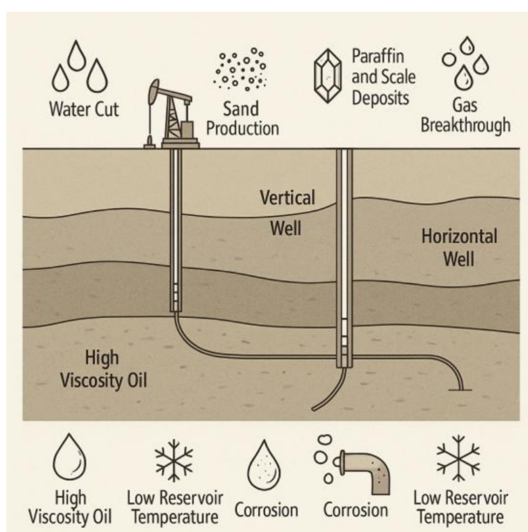

Figure 2—List of Complicating Factors in Production.

A virtual flowmeter makes it possible to reconstruct the dynamic behavior of well production based on indirect parameters — such as pressure data and telemetry from the control station — with much higher frequency [8]. The objective of this study is to develop and validate an integrated hydraulic-electrical model of a virtual flow meter for wells equipped with ESP. Unlike existing methods that are adapted solely to hydraulic parameters or are purely data-oriented ML approaches, the proposed algorithm ensures physical consistency, takes into account pump degradation and electrical losses, and provides a real-time flow estimate with an error of less than 1%.

Related work and theory

Physical models

Physico-mathematical modeling of multiphase flows in a well forms the foundation for virtual flow metering models. These models describe fluid flow processes in the well with sufficient accuracy and are widely used in the industry [9]. The model includes the following components: a Black Oil fluid model [10], wellbore inclination data, temperature distribution, a model of the tubing and casing, a model of the electric submersible pump (ESP) [5], [11], [12], [13], and, if applicable, a choke model [14], [15], [16]. Physical model for virtual flow meter based on building pressure distribution along the wellbore [17], [18], [19].

A physical model is an approximation, which may deviate from real-world data; therefore, any such model requires adaptation. Many researches presents [6], [7], [13], [20] a virtual flowmeter model that is adapted using the inlet pressure of the pump [12] (ESP models).

Machine-learning and neural network models

Machine learning-based virtual flowmeter models are an alternative to traditional physico-mathematical models [21].

The core principle of an ML-based model is to identify relationships between input and output parameters obtained from the field. These models do not consider the physical nature of the system and do not require physical or analytical calculations of oil and gas production processes. On the one hand, this accelerates the computation of the virtual flowmeter; on the other hand, such a model functions as a "black box" — meaning it is not explicitly clear what is happening inside the model [22]. It is not possible to directly access the internal mechanisms of the model to understand which physical processes are occurring in the well, making it more difficult to monitor operations in detail and perform thorough data analysis.

Various machine learning algorithms can be applied to virtual flow metering, such as:

- Linear Regression (LR) [23],
- Gradient Boosting (GB) [21],
- Support Vector Machine (SVM) [24],
- Long short-term memory (LSTM) [25].

Hybrid models

Hybrid models include physical models and machine-learning models. One type of hybrid modeling involves the use of proxy models, where the physical model is employed to generate input features that are then analyzed by machine learning algorithms [12], [22], [26].

This approach proves especially beneficial when data availability is limited or the data is noisy, which is often the case in real-world oilfield operations. For example, in the event of sensor malfunctions, the physical model can fill in the gaps by producing approximated yet plausible values [27].

However, the hybrid approach suffers from some problems inherited from machine learning models, such as low interpretability of results and the need for a large amount of data. It is also worth noting that when data scarcity is critical, relying solely on purely physical models may offer more robust and reliable results [22].

Alternative hybridization strategies propose a combination of multiple models, where some are trained directly on raw field data, while others are trained on features generated by the physical model [7]. The final flow rate prediction is then computed as a weighted sum of the outputs from both types of models.

Physical models are highly interpretable but require careful calibration, while machine learning models provide fast computational processing but lack transparency and depend on large volumes of data. Hybrid models partially address these issues but still do not take into account the electrical performance of the ESP. This gap served as the basis for developing the integrated approach presented in this paper.

Methodology overview

Required data

As an alternative, this paper proposes a comprehensive model of a well equipped with an ESP, which accounts for both hydraulic and electrical parameters.

This paper [28], [29] presents an algorithm that uses both high-frequency data from telemetry sensors and static data. Combined model needs such parameters as telemetry sensors data (intake pressure, frequency, amperage, etc.) [30], well characteristics, electric submersible pump parameters, PVT properties of the oil fluid [10] and others. The flow rate can be measured by automatic group flow measuring system (AGFMS) or multiphase measuring system. Also, for adaptation, the flow rate measured with the AGFMS is required.

Table 1—Features used for physical virtual flow meter model.

Data category	Parameters	Description
Static data	Electric submersible pump data, motor description, gas separator data, casing and tubing characteristics, true vertical and measured depth of well, PVT properties	Rarely measured static data
Telemetry data	Intake pressure, linear pressure, annular pressure, ESP frequency, motor load, amperage, control station power	Dynamic data from telemetry sensors

Physical model conception

A physical model is an approximation, which may deviate from real data; therefore, any such model requires adaptation. The paper [20], [29] presents a virtual flowmeter model that is adapted based on the inlet pressure of the pump.

The calculation algorithm of the physical model for a well equipped with an electric submersible pump (ESP) includes the following steps [11]:

1. Pressure distribution along the casing is calculated, and the inlet pressure of the ESP is determined.
2. The natural, artificial, and total separation coefficients at the pump inlet are computed.
3. The PVT property correlations are adjusted, taking separation into account.
4. The pressure distribution within the pump is calculated, and the ESP discharge pressure is determined.
5. The pressure distribution along the tubing is calculated, and the buffer pressure is determined.
6. If a choke is present, the pressure drop across the choke is calculated, and the wellhead (linear) pressure is determined.

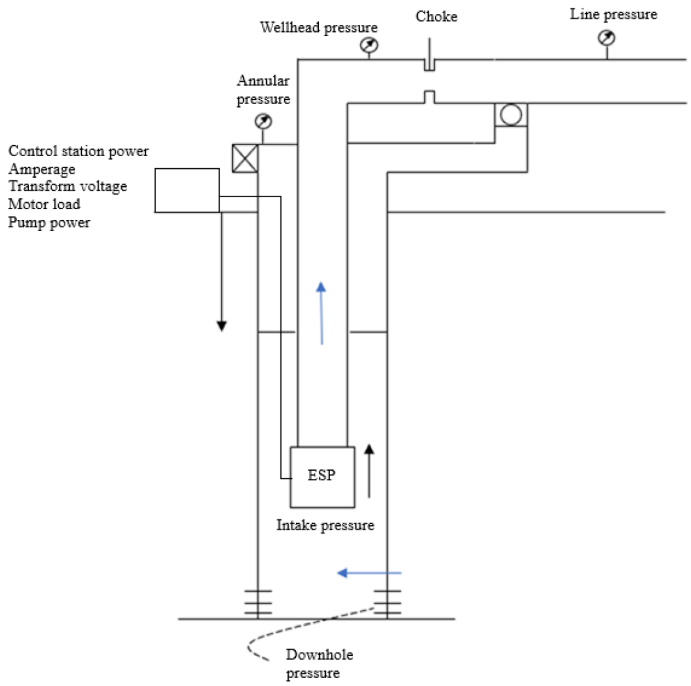

Figure 3—Design of a Producing Well Equipped with an ESP.

To improve the accuracy of the model, the paper presents a comprehensive model that takes into account electrical parameters, since adaptation only to the pressure at the intake does not take into account the loss of electricity, power and other parameters, in the entire ESP system and cannot fully account for pump degradation [29].

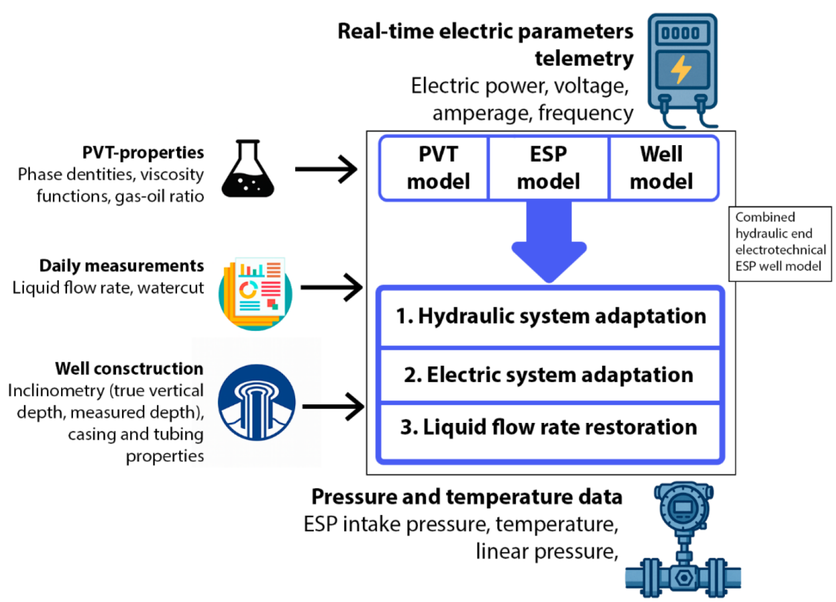

Figure 4—Schematic block diagram of combined ESP well model, presented in this paper: PVT-properties [31], daily measurements, telemetry, and well design data are received from the field and laboratory studies. A black oil fluid model, an ESP model based on the passport characteristics, and a model of the well itself are created. Then the algorithm works in three stages: 1. Adaptation of the hydraulic system, 2. Adaptation of the electrical system, 3. Recovery of the flow rate. All this constitutes a combined model of the ESP well.

Model Adaptation

As in many existing approaches, the first step in solving the virtual flow metering problem is model adaptation. In this approach, adaptation should be divided into three key stages [20]:

- **Step 1:** Hydraulic system adaptation;
- **Step 2:** Electrical system adaptation;
- **Step 3:** Selection of optimization function coefficients.

Hydraulic System Adaptation

Hydraulic calculation adaptation is performed by minimizing the error based on the intake pressure of the ESP or, if this parameter is unavailable, based on the dynamic fluid level in well [17], [18].

1. **Pump degradation coefficient calculation**. This accounts for discrepancies between the actual intake pressure of the pump and the calculated value.

$$\varepsilon = \left| p_{in_{calc}}(h_{degr}) - p_{in_{meas}} \right| \tag{1}$$

$$dp_{stage} = \rho_{fluid} H_{\mathrm{corr}} g * head\ factor \tag{2}$$

2. Calculation of the pump degradation coefficient can also be carried out by adapting to the dynamic level.

$$\varepsilon = \left| h_{dyn_{calc}}(h_{degr}) - h_{dyn_{meas}} \right| \tag{3}$$

By minimizing the error function, the pump degradation coefficient is calculated, which acts as a multiplier in determining the ESP head.

Electrical System Adaptation

After completing the hydraulic adaptation, the next step is adapting the electrical part. This adaptation is based on actual field data:

- Actual pump power (W_{pump});
- Actual amperage of the ESP motor (I_{meas});
- Actual ESP motor load ($Load_{meas}$);
- Actual voltage at the transformer tap (U_{surf_meas});
- Actual active power at the control station ($W_{cs\ meas}$).

Adaptation is performed by calculating adaptation coefficients for these parameters. For correct execution, at least one of the above actual parameters must be available. In a comprehensive well model calculation with an ESP, the following adaptation coefficients are determined [20]:

1. **Pump power adaptation coefficient** calculation (C_{pump}):

$$\varepsilon = |I_{calc}(C_{pump}) - I_{meas}| \tag{4}$$

$$W_{pump} = W_{pump\ calculated} * C_{pump\ power} \tag{5}$$

where:

- ε – error function;
- I_{calc} – calculated ESP motor amperage, A;

- I_{meas} – actual measured ESP motor amperage, A;

2. **Amperage correction coefficient** calculation, accounting for discrepancies between the calculated ESP motor characteristic and the actual one (C_I):

$$\varepsilon = |I_{calc}(C_I) - I_{meas}| \tag{6}$$

$$I = I_{nom} * \frac{\%Amp}{100\%} * C_I \tag{7}$$

where:

- I_{meas} – actual measured ESP motor amperage, A;
- I_{nom} – nominal ESP motor amperage, A;
- $\%Amp$ – percent ESP motor amperage determined from the motor characteristic at motor loading $load_{meas}$, %. If actual load is not specified, the calculated value is used.

3. **ESP motor load adaptation coefficient** calculation, accounting for discrepancies between calculated and actual ESP load (C_{load}):

$$\varepsilon = ||Load_{meas} - Load_{calc}(C_{load})|| \tag{8}$$

$$Load = \frac{I * f}{I_{nom} * f_{nom}} * C_{Load} \tag{9}$$

where:

- $Load_{meas}$ – actual measured ESP motor load, %;
- $Load_{calc}$ – calculated ESP motor load, %;
- I – re-calculated ESP motor amperage from equation (7), A.
- f – actual measured ESP shaft rotation frequency, Hz;
- f_{nom} – nominal ESP shaft rotation frequency, Hz;

4. **Voltage adaptation coefficient at the transformer tap**, accounting for discrepancies between the transformer tap voltage and the ESP motor voltage ($C_{transform\ voltage}$):

$$\varepsilon = |U_{calc}(C_{transform\ voltage}) - U_{meas}| \tag{10}$$

$$U_{transform} = (U_{motor} + U_{cable\ loss}) * C_{transform\ voltage} \tag{11}$$

ESP motor voltage based on the known voltage at the control station:

$$U_{motor} = U_{surf_{calc}} - 1,732 * I_{calc} * R_{cable} \tag{12}$$

where:

- U_{calc} – calculated voltage, V;
- U_{meas} – actual voltage, V;
- U_{motor} – motor voltage, V;
- $U_{surf_{calc}}$ – required surface voltage, V;
- $U_{cable\ loss}$ – voltage drop in the cable, V;
- I_{calc} – calculated ESP motor amperage, A.
- R_{cable} – cable resistance, ohms.

5. **Control station power adaptation coefficient** calculation, accounting for discrepancies between the calculated ESP motor power and the measured control station power (C_{cs}):

$$\varepsilon = |W_{cscalc}(C_{cspower}) - W_{csmeas}| \quad (13)$$

$$W_{cs} = W_{cable} C_{cspower}(2 - \eta_{cs}) \quad (14)$$

where:

- $W_{cs\ calc}$ – calculated active power at the control station, W;
- $W_{cs\ meas}$ – measured active power at the control station, W;
- W_{cable} – active power at the cable input, W;
- η_{cs} – control station efficiency, dimensionless.

In Figure 5, the calculated values of the above-described calibration coefficients for various measurements are presented. Based on these results, it can be concluded that the model exhibits a high degree of physical consistency, as all key calibration coefficients fall within acceptable ranges around unity. Figure 6 shows the dynamics of all field parameters to which the adaptation described in this chapter was applied. It is worth noting that the dynamics of the parameters differ significantly in periodicity and trends. Some parameters remain constant for a long time. This will subsequently affect the choice of scales for the flow rate recovery formula, which will take into account electrical and hydraulic parameters. The adaptation coefficient for the voltage at the transformer tap changes slightly, since the parameter itself often remains constant according to telemetry data, but the calculated voltage value may differ slightly from the measured one.

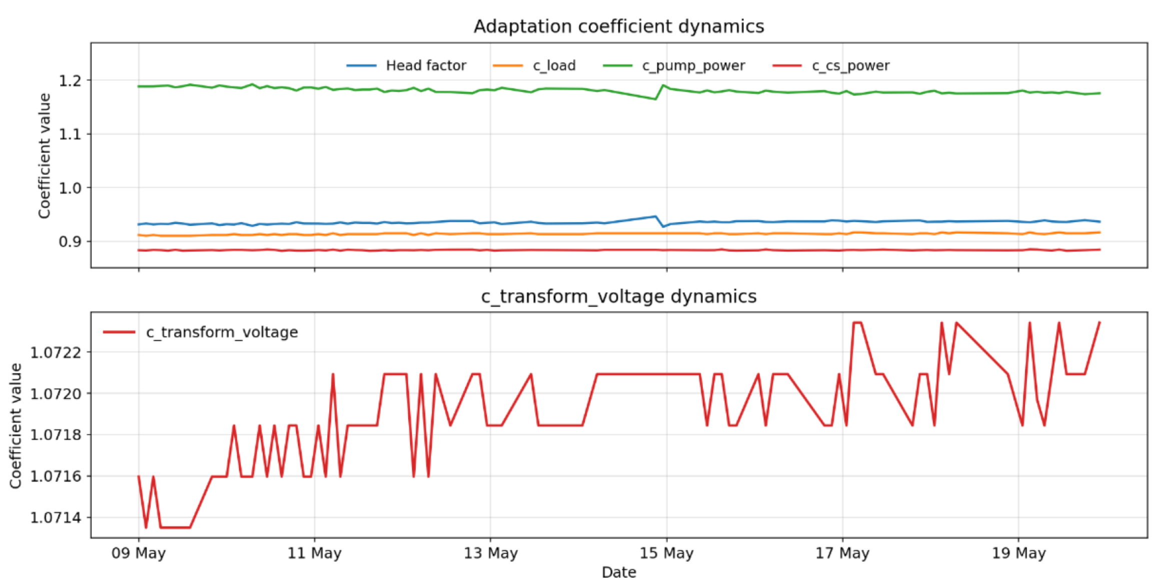

Figure 5—Adaptation coefficients dynamics.

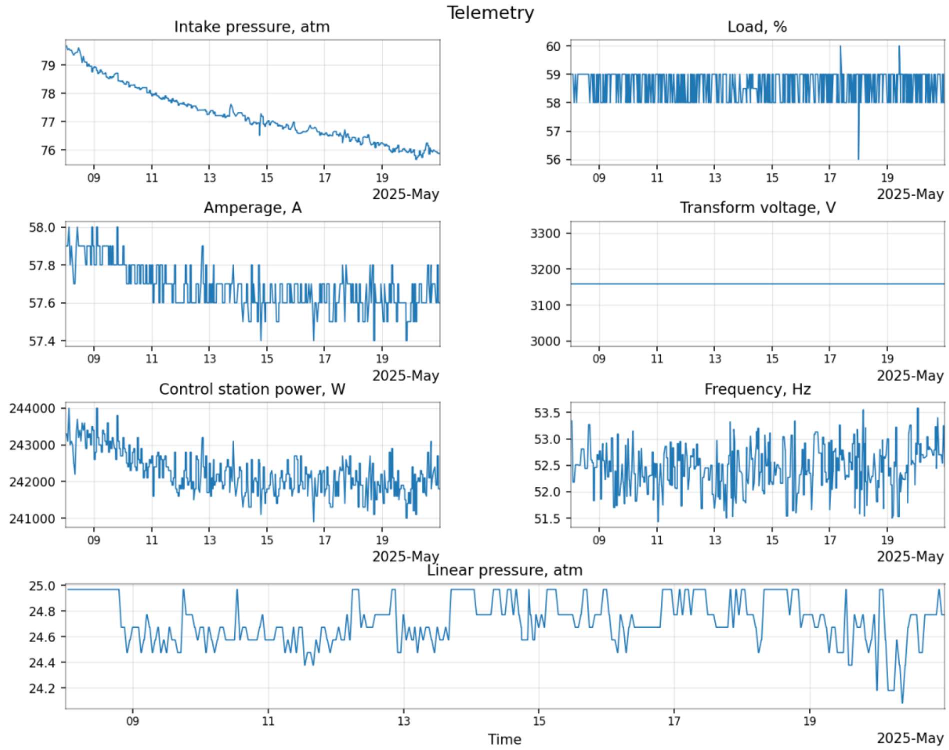

Figure 6—Telemetry used for ESP well adaptation.

The figure 7 shows the distribution of adaptation coefficients obtained during large-scale validation. They were obtained by minimizing the error, with boundaries from 0 to 3. Figure 7 shows that the pump degradation factor varies from 0,8608 to 1,4644 (by 95% range), as it requires very fine tuning, and the slightest change in this factor, even within thousandths, makes a significant contribution to the final recovery of the flow rate. The load varies from 0,6760 to 2,3026 (by 95% range) within wide limits, as it can sometimes change significantly within one month. The pump power factor is closest to one (median value is 1,0062), as the calculations and measured values for this parameter often converge. The voltage at the tap often has a value greater than one (varies from 1,1786 to 1,3968 by interquartile range), as the voltage parameter itself does not change often, but with a combined calculation, discrepancies with the measured parameter are obtained, so the calculation needs such a large adjustment in the form of a factor greater than one. The power factor at the control station often lies in the range from 0,6689 to 0,7981 by interquartile range.

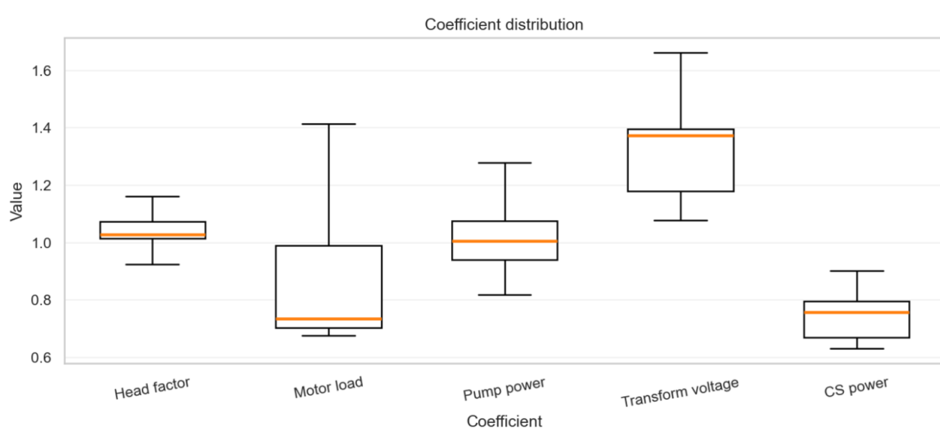

Figure 7—Adaptation coefficients distribution.

After determining all adaptation coefficients and substituting them into the direct calculation, the actual measured parameters should match the calculated ones, provided the adaptation was correct and the coefficients did not reach their limits (zero or maximum permissible values).

Weight Selection in the Optimization Problem

A distinctive feature of this approach is additional weight calibration. This is necessary to solve the optimization problem of production rate recovery while minimizing error across all calibrated parameters in the model. The following equation (15) is used:

$$\frac{\varepsilon = \sqrt{w_1 * \left(\frac{p_{in\ calc} - p_{in\ meas}}{p_{in\ meas}}\right)^2 + w_2 * \left(\frac{Load_{calc} - Load_{meas}}{Load_{meas}}\right)^2 +}}{+ w_3 * \left(\frac{I_{calc} - I_{meas}}{I_{meas}}\right)^2 + w_4 * \left(\frac{U_{calc} - U_{meas}}{U_{meas}}\right)^2 + w_5 * \left(\frac{W_{cs\ calc} - W_{cs\ meas}}{W_{cs\ meas}}\right)^2} \tag{15}$$

where:

w_1, w_2, w_3, w_4, w_5 – weights for the terms of the error function.

In the basic case, this task can be solved by assigning equal coefficients to all calibration coefficients in the model. However, research has shown the need for an additional adaptation stage due to the varying impact of calibration coefficients for different parameters (e.g., hydraulic vs. electrical parameters) on adaptation results.

Figure 8 shows correlation between adaptation coefficients and restored flow rate after large-scale validation. using wells data in Western Siberia, demonstrating how accounting for specific parameters during model adaptation affects the error in production rate recovery compared to the original measurements. This is the first stage of the study of the parameters used for adaptation. At this stage, it is possible to estimate which parameters contribute most to the error.

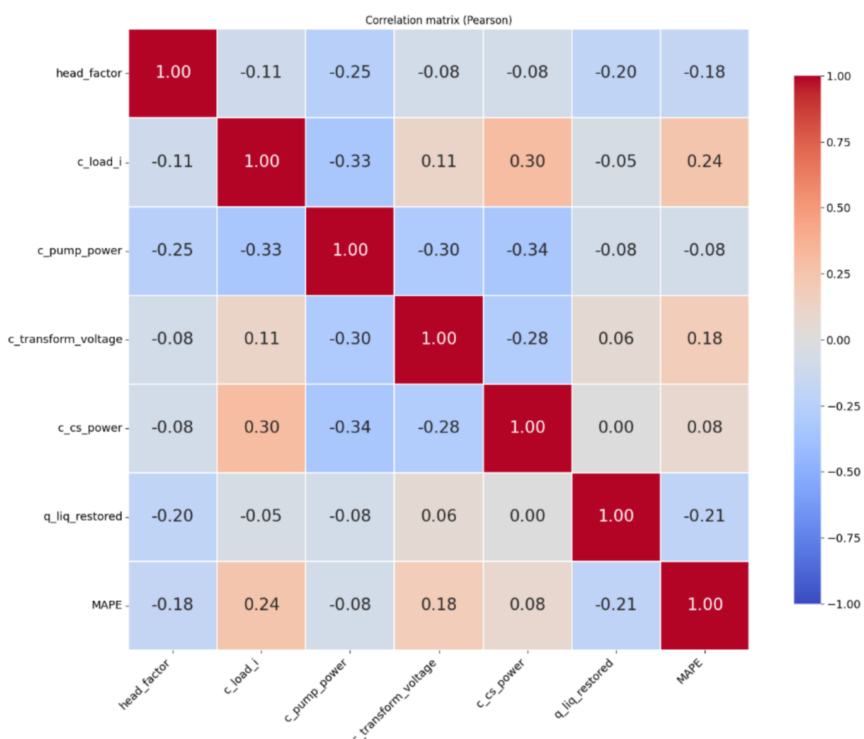

Figure 8—Correlation Matrix (Pearson) of adaptation coefficients.

During mass testing on different wells with ESP, by minimizing error between measured flow rate and calculated flow rate [32] by using different weights (second step of mass testing), the research of parameters showed that the weights in general should be $w_1 = 1$, $w_2 = 0.05$, $w_3 = 0.55$, $w_4 = 0.2$, $w_5 = 1$. With such weights, the virtual flow meter based on the combined model shows the best results in terms of accuracy.

The figure 9 shows the flow rate calculations using different parameters and different weights.

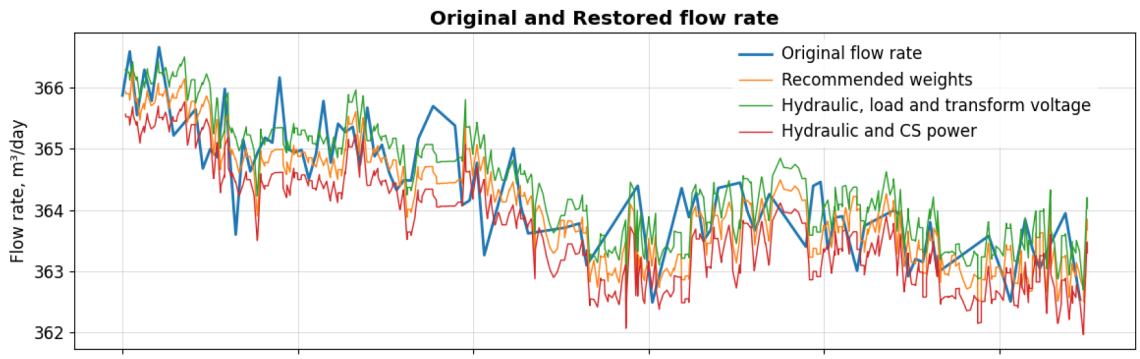

Figure 9—Restored rate using different weights and parameters for adaptation.

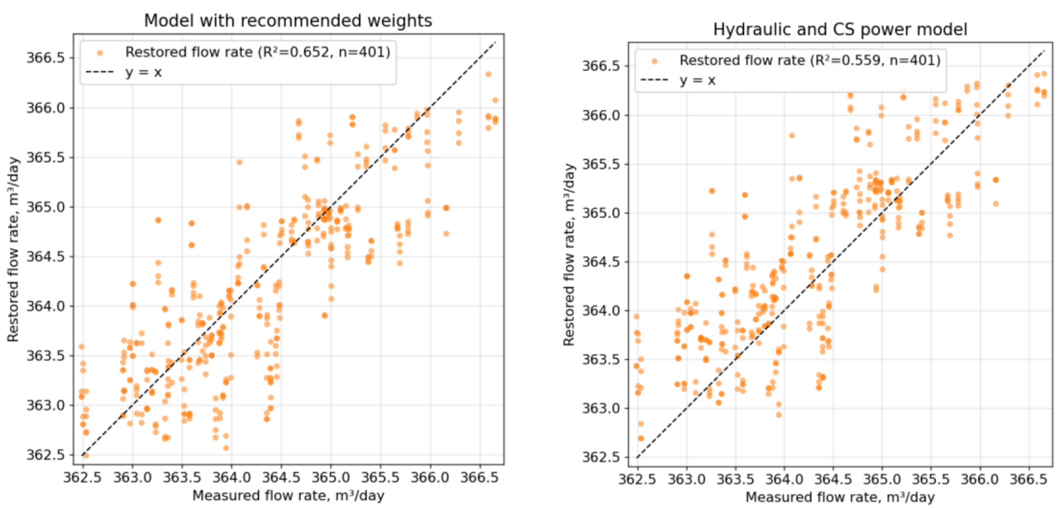

Figure 10—Model with recommended weights and model with hydraulic and CS power adaptation comparison.

The research showed that the most important parameter is the pressure at the pump intake, that is, the adaptation of the hydraulic system. Figure 11 shows the error distribution on a single well. Figure 12 shows the results of mass testing, where the metrics are worse than in Figure 11, as this is explained by the large sample, but it is more clearly seen that the approach with the recommended weights shows itself better, compared to other approaches. The median error for the model with the recommended weights is the smallest compared to other combinations and is 4,866%. Without adaptation of the hydraulic system, using only electrical calculations, the median error reaches up to ~30% compared to the original flow rate (figure 13). Transform voltage changes infrequently, so its contribution to the accuracy of the formula is minimal. Amperage is sensitive to harmonics, short-term peaks, network imperfections; in the objective function it does not always make the model more accurate. While the control station power behaves more stably and makes a significant contribution to the accuracy of the model. Motor load reflects the mechanical part, the connection between electrical and hydraulic systems, and it greatly increases the error when taking this parameter into account. Its weight is equal to 0.05.

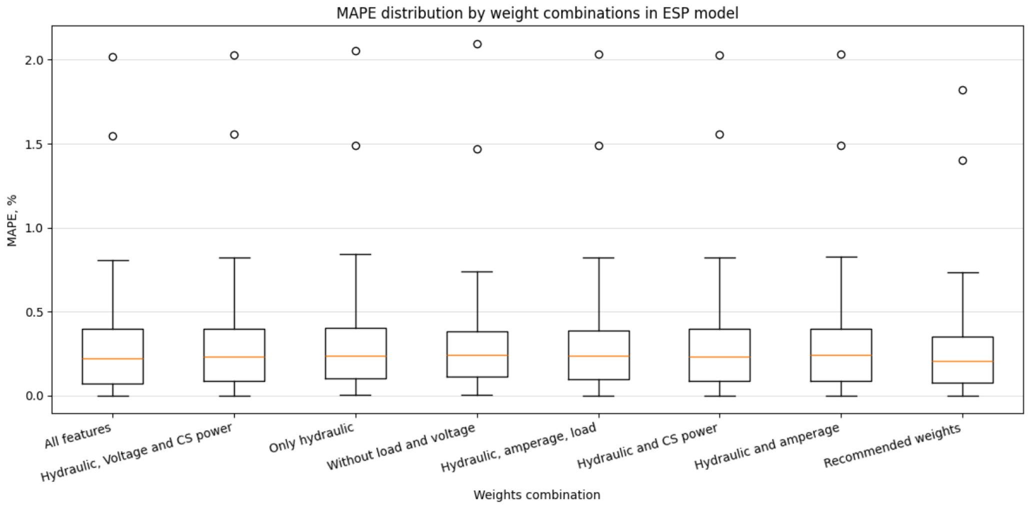

Figure 11—MAPE distribution by weight combinations (one well).

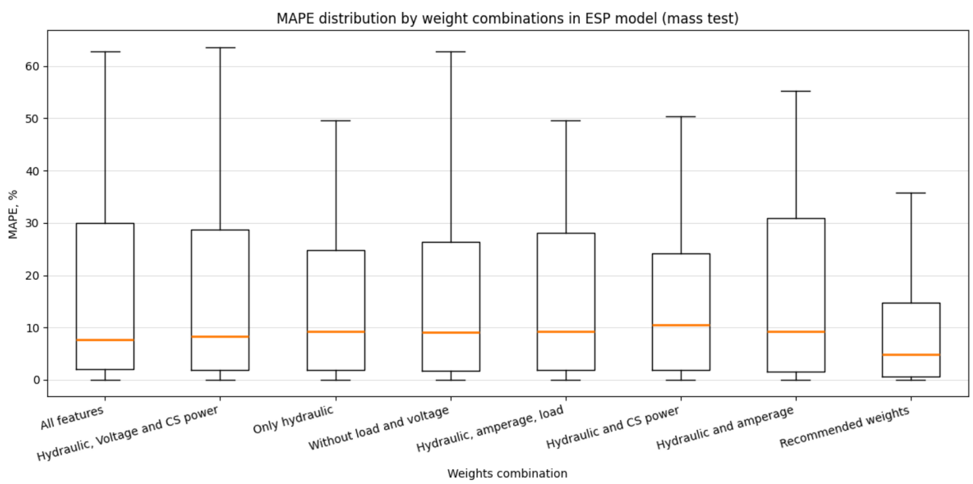

Figure 12—MAPE distribution by weight combinations (mass test on different ESP wells).

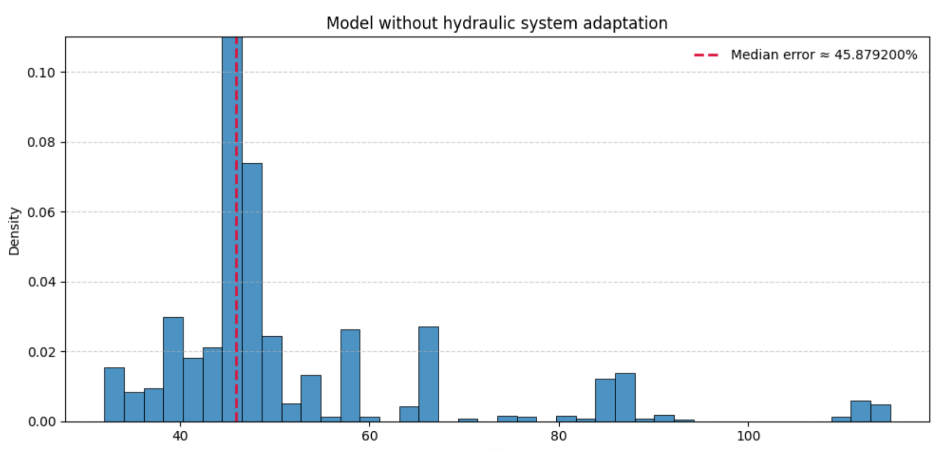

Figure 13—Model without hydraulic system adaptation error.

It is important to note the flexibility of the combined model of a well with an ESP. If some parameter is poorly measured, there are inaccuracies in the initial data on electrical engineering, then the weights can be adjusted, making the model more accurate. Therefore, the weights presented in the article are, in general, a universal option, however, if necessary, they can always be adjusted.

Figure 14 shows difference between model with electric and hydraulic system adaptation and model with only hydraulic system adaptation, after large-scale validation. The median error in the model using only the hydraulic system is 1,549% higher, indicating that when using additional electrical parameters for adaptation, the model is better able to solve the virtual flow metering problem.

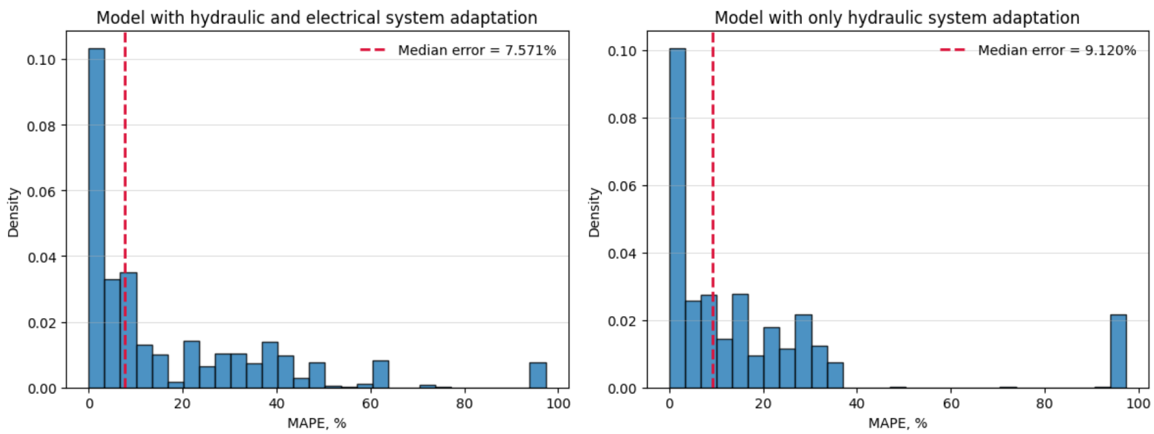

Figure 14—Difference between model with hydraulic + electrical system and only hydraulic system adaptation.

Flow Rate Recovery

Once the adaptation coefficients are obtained, production rate recovery can begin. Coefficients that cannot be calculated for technical reasons are set to unity.

To recover the production rate, the following parameters are required: line pressure, water cut, ESP intake pressure, control station power, ESP motor amperage, ESP motor load, transformer tap voltage, and current frequency. A similar function is used for recovery, but with the selected coefficients:

$$\varepsilon = \frac{\sqrt{\left(\frac{p_{in\,calc}-p_{in\,meas}}{p_{in\,meas}}\right)^2 + 0,05 * \left(\frac{Load_{calc}-Load_{meas}}{Load_{meas}}\right)^2 +}}{+0,55 * \left(\frac{I_{calc}-I_{meas}}{I_{meas}}\right)^2 + 0,2 * \left(\frac{U_{calc}-U_{meas}}{U_{meas}}\right)^2 + \left(\frac{W_{cs\,calc}-W_{cs\,meas}}{W_{cs\,meas}}\right)^2 \rightarrow min} \quad (16)$$

Thus, knowing the adaptation coefficients and actual ESP well parameters, production rate is recovered through physical and mathematical calculations. As an example of production rate recovery, With the current method, it is possible to achieve an error of less than one percent when comparing the recovered production rate with actual measurements.

The figure 15 shows an example of the error function after adaptation for calculating the flow rate and its minimum (177,25 $\frac{m^3}{day}$). Using the Brent's method, the minimum of the error function is found by iterating over possible liquid production rates. For each rate, ESP well parameters are recalculated, and the scalar function minimum is sought over a given interval. This method combines the golden section search with parabolic interpolation for predicting the minimum [32].

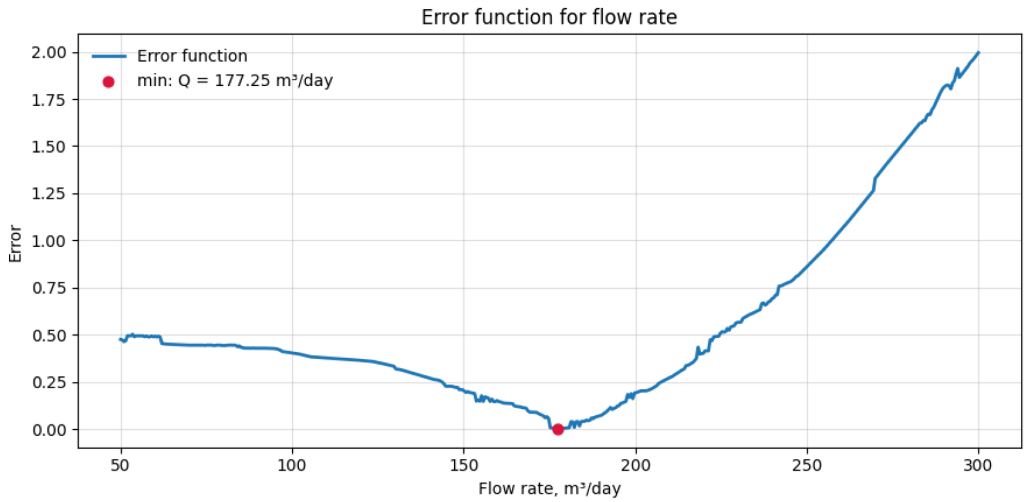

Figure 15—Error function for combined VFM ESP model.

Results

In this work, a study was conducted to assess the impact of accounting for various coefficients on the quality of production rate recovery in wells. The results of this study showed that, in addition to the mandatory head correction factor, which accounts for the hydraulic component and is essential for solving the virtual flow metering problem, it is critically important in the electrical part of the calculation to perform adaptation for all key parameters described in this article. Average adaptation time is 1,2 seconds. Average flow rate recovery time is 0,7 seconds.

Figure 16 also presents an example of production rate recovery using different approaches. The hybrid and ML models deviate quite strongly from the measured flow rate in some places, while the model presented in the article more accurately follows the trends in the measured flow rate, since it takes into account more physical and electrical parameters. Figure 17 presents mean absolute percentage error between measured flow rate and calculated flow rate, by using different models.

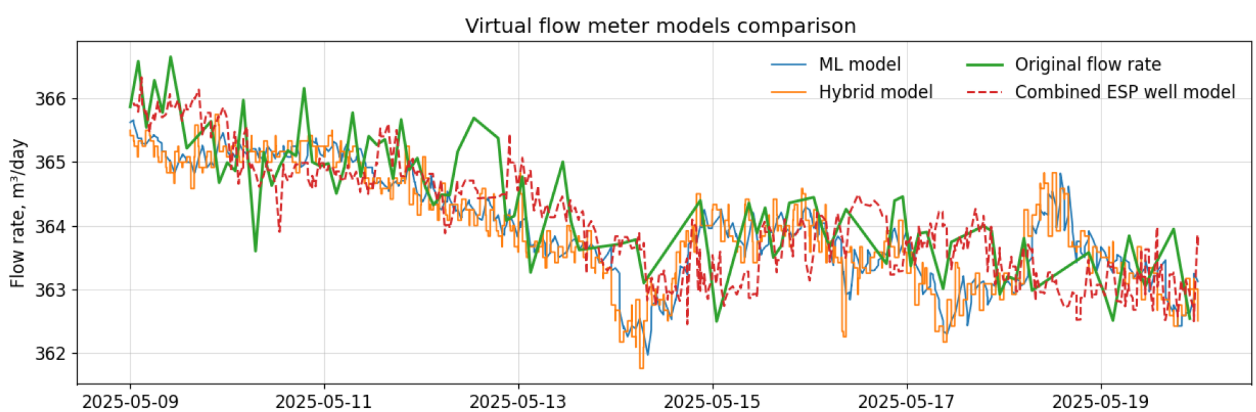

Figure 16—Virtual flow meter models comparison: original flow rate, Machine-Learning model, Hybrid model and Combined ESP well model.

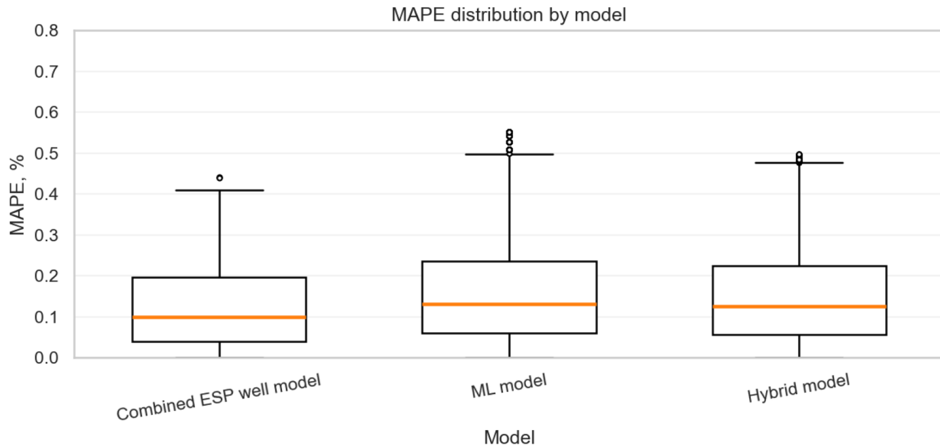

Figure 17—MAPE distribution by different models: Combined ESP well model, Machine-Learning model, Hybrid model.

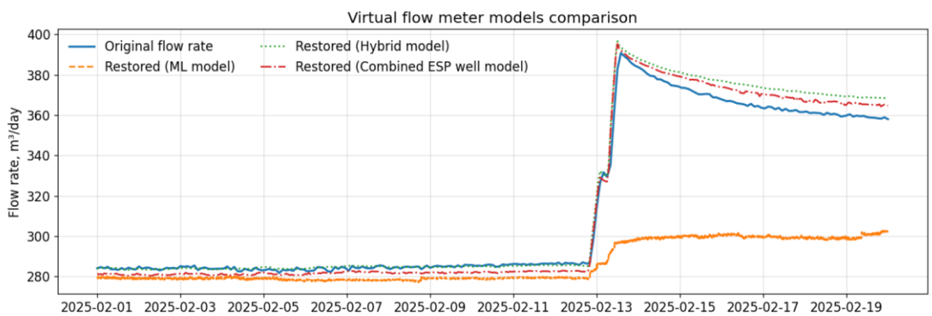

Figure 18—Virtual flow meter models comparison: Case when the operating mode at a well changes; original flow rate, Machine-Learning model, Hybrid model and Combined ESP well model.

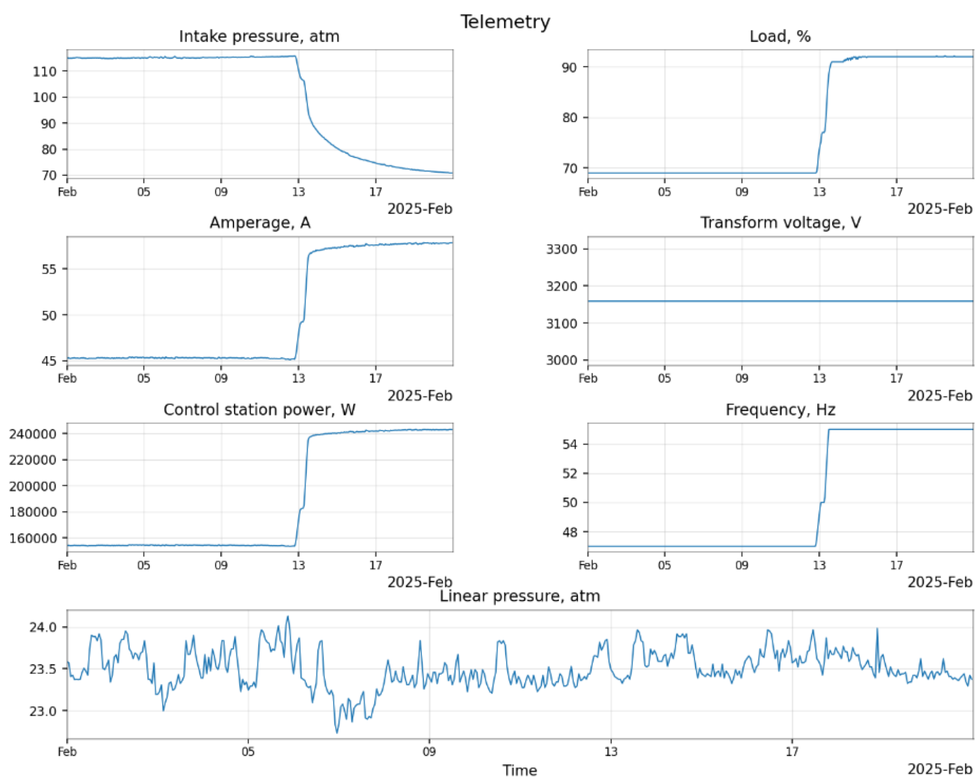

Figure 19—Telemetry used for ESP well adaptation in case when the operating mode at a well changes.

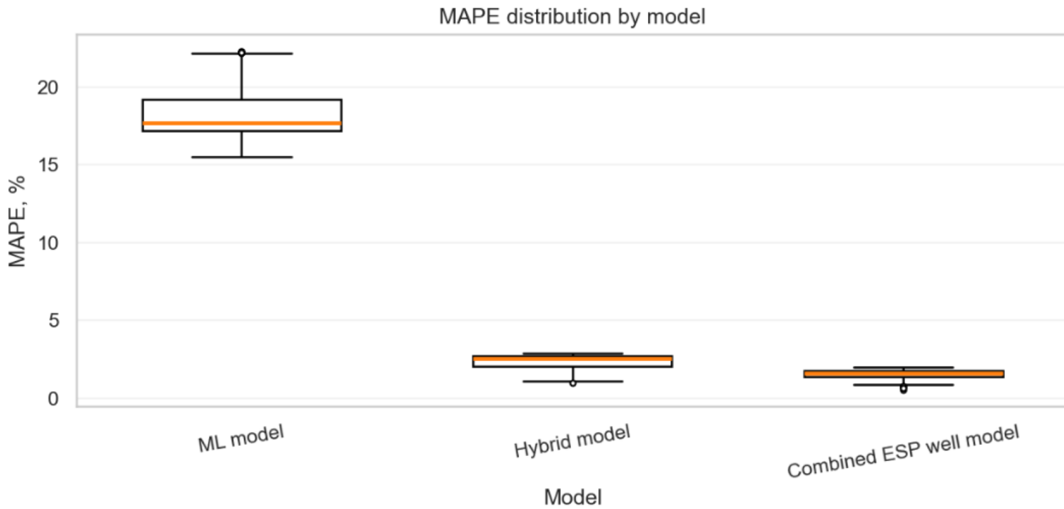

Figure 20—MAPE distribution by different models: Case when the operating mode at a well changes; original flow rate, Machine-Learning model, Hybrid model and Combined ESP well mode.

The figures 16–17 shows a comparison of the model presented in this article with the ML model and the hybrid model [21], [33]. As the study showed, taking into account the hydraulic system together with the electrical parameters gives more accurate results in restoring the flow rate than other models. The average MAPE of the integrated ESP well model is 0.129%. The hybrid model showed an average error of 0.153%. The pure ML model, which does not use physics interpretation, showed an average error of 0.161%. Thus, the increase in accuracy compared to the hybrid model was 1.18%. Figures 18–20 demonstrate the results of the models when the well operating mode changes. In this case, adaptation was carried out before the mode change, and subsequent recovery. As the results showed, ML, due to the fact that it does not take physics into account, was unable to accurately predict the flow rate, and gave an average error of 7.98%. At the same time, the combined and hybrid models were able to restore the flow rate, and took into account the change in mode. The combined model showed an average error of 0.235% better.

Table 2—Comparison of the percentage (%) error between the measured flow rate and the reconstructed flow rate for different models.

Case number	ML model	Hybrid model	Combined ESP well model
Case 1	1.904	0.403	1.374
Case 2	22.477	3.643	3.307
Case 3	19.288	2.039	1.501
Case 4	17.782	2.623	1.725
Case 5	16.676	2.816	1.813
Case 6	14.970	2.892	1.922
Case 7	15.577	3.093	2.157
Case 8	16.454	3.263	2.304

Model Limitations and Applicability

The proposed integrated model demonstrates high accuracy (<1 % MAPE under validation), yet several practical constraints must be considered for reliable field deployment.

Data and sensor requirements

Model initialization strictly requires ESP shaft frequency and intake pressure (or dynamic fluid level), as well as periodic liquid flow-rate measurements for proper calibration. For higher accuracy, high-frequency telemetry of active power, current, and voltage is recommended. Significant sensor noise or frequent data gaps necessitate robust preprocessing and may reduce performance.

Computational load

The two-stage adaptation and iterative flow-rate optimization (e.g. Brent's method) are computationally more demanding than purely data-driven approaches. Real-time deployment for large well portfolios requires adequate processing capacity.

Operational boundaries

- Flow regime sensitivity: the model performs best under steady or slowly varying conditions; during rapid transients (severe slugging, abrupt ESP shutdown/start-up) accuracy may temporarily degrade.
- Calibration needs: periodic adjustment against reference flow-rate measurements (e.g. test separators) is required, with frequency increasing in unstable wells.
- Physical assumptions: accuracy is limited by underlying multiphase-flow correlations, which may be less reliable for extreme fluid behaviors (e.g. severe emulsions, foam flow).

Optimal and non-optimal use cases

The model provides the greatest benefit where simpler approaches fail — wells with high GOR, gas interference, progressive pump wear, or fluctuating electrical supply. Conversely, in wells lacking reliable intake-pressure data or experiencing rapid, unpredictable transients, simpler data-driven methods may be more practical.

In summary, optimal results are achieved when reliable frequency and intake-pressure data, periodic flow-rate measurements, and sufficient processing power are available, and the model is applied in conditions where its integrated hydraulic-electrical approach offers a clear advantage.

Conclusion

This study developed and field-validated a comprehensive VFM algorithm for ESP-lifted wells that jointly adapts hydraulic and electrical models. Large-scale testing confirmed that the integrated approach consistently outperforms classical hydraulic-only and hybrid methods, reducing flow rate estimation error to below 5%.

By incorporating real-time electrical telemetry alongside hydraulic parameters, the model better reflects actual well conditions, improves production monitoring accuracy, and reduces reliance on costly surface or downhole test equipment. The approach is particularly beneficial for mature fields with infrastructure limitations and complex operating conditions.

Future enhancements include embedding a machine-learning core into the current physically consistent framework to further refine coefficient calibration under highly variable production regimes. Overall, the integrated model provides a robust and scalable solution for optimizing ESP performance and minimizing production uncertainty.

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